

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Spring- 2026

Course Title: Concepts of Moral Development

Course Code: MDP-3503, 5th Semester

Full Marks : 50

Time: 2.50 Hours

[Figures on the right hand margin indicate full marks]

Group-A

(Answer any 2 of the following questions)

Q.1

- a) Define Morality and Values 2
- b) What is the basis of morality in Islam? 2
- c) Figure out the causes of moral decay and its remedial 6

Q.2

- a) What is the significance of Islamic dress code? 2
- b) Mention general characteristics of Islamic dress. 3
- c) Discuss the influence of Islamic dress in the formation of a society adorned with morality. 5

Q.3 : Write short notes on:

- a) Rights of the parents and children 2
- b) Health and hygiene in the light Islam. 2
- c) Social disaster caused by drug and gambling? 6

Group-B

(Answer any 3 of the following questions)

Q.4

- a) Why 'July Revolution 24' was inevitable? 2
- b) Determine the contribution of Gen Z in 'July Revolution 2024' 2
- c) What are the socio-political changes you dream after the 'July Revolution 2024'? 6

Q.5

- a) In what ways students can play a pivotal role in the formation of an ideal society? 5
- b) Explain the importance of Halal earnings and professions from Islamic perspective. 5

Q.6

- a) Determine certain salient features of Islamic Culture and Western Culture. 2
- b) Explain the Islamic & Western concepts of entertainment. 2
- c) Elucidate Shariah ruling on music, literature and cinematography. 6

Q.7

- a) Discuss the 7 categories of people who will get shelter under the shadow of Arsh on the Day of Judgment. 5
- b) Explain the 5 questions that will be asked on the day of Resurrection. 5

International Islamic University Chittagong

Center for General Education (CGED)

Semester End Examination- Spring -2026

Course Code: URED-3503(URED-3101 for LLB)

Course Title: Political Thoughts and Social Behavior

Time: 2:30 Hours

Full Marks: 50

Answer the following questions

(All questions are of equal value)

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	Discuss the objectives and significance of marriage in Islam. Or Discuss the causes of matrimonial disputes and their impact on family stability.	10 10	3	Remember
2	Analyze the status of women before and after the era of Prophet Mohammad (SAAS).	10	3	Analyze
3	Develop and justify a comprehensive framework for Islamic parenting that integrate spiritual, ethical and emotional development addressing contemporary societal challenges.	10	3	Evaluate
4	Evaluate the effectiveness of the Islamic prohibition of Ribā in promoting economic justice and social welfare.	10	3	Evaluate
5	"Islamic law balances freedom and moral responsibility by permitting all lawful things while prohibiting both unlawful acts and the means leading to them." Analyze	10	3	Analyze & Evaluate

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Final Examination, Spring 2026

Course Code: CSE-3529

Course Title: System Analysis and Design

Total marks: 50 Time: 2 hours 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks]

GROUP: A

1. a) A banking system manages customer accounts, deposits, withdrawals, and transactions. Customers can check account balances and perform transactions. Bank employees manage customer records. The system generates transaction reports. Based on the above scenario, 5 CO3
- i) Draw a Context Diagram.
- ii) Draw a Level 1 DFD with the following processes:
- Account Management
 - Transaction Processing
 - Report Generation

OR (of 1a)

A bank loan approval system decides whether a customer can get a loan based on several conditions: 5 CO3

1. If customer has a good credit score, stable income, no previous loan default, and requested amount is within limit → **Approve Loan**
2. If customer has a low credit score → **Reject Loan**
3. If customer income is insufficient → **Request Additional Documents**
4. If customer has previous loan defaults → **Reject Loan**
5. If loan amount exceeds the maximum limit → **Reduce Loan Amount**

Based on the above scenario

- a) Draw a decision tree showing how the system reaches a decision.
b) Create a decision table showing all possible combinations of conditions and outcomes.
1. b) Write short notes on: Data Dictionary, Structured English, Pseudocode, Decision Trees and Decision Tables 5 CO1
2. a) Differentiate between the Top-Down and Bottom-Up strategies of system design, using the modular approach as the basis of comparison. 5 CO2

OR (of 2a)

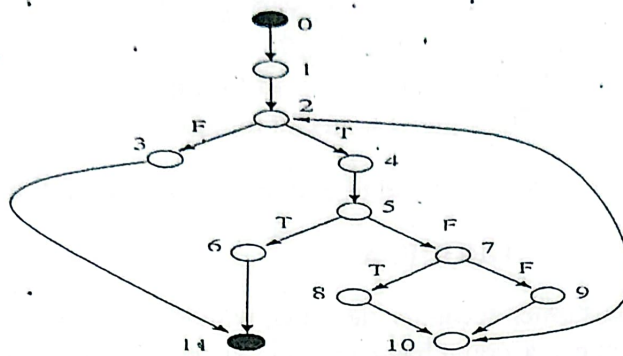
What is input design? What are the main sections recommended for good form design?

- b) Distinguish between a conceptual data model, a logical data model, and a physical data model, giving one feature unique to each. 5 CO2

GROUP: B

3. a)

5 C02



- I. What is cyclomatic complexity? Why is cyclomatic complexity important in white-box testing?
- II. Calculate the cyclomatic complexity of the CFG.

- b) Explain three differences between Black Box and White Box Testing. Choose the most appropriate testing approach (**Black Box Testing or White Box Testing**) for each scenario and justify your answer:

5 C02

- a) Checking whether a registration form accepts valid and invalid inputs.
- b) Testing all branches of an employee salary calculation program.
- c) Verifying whether a mobile application meets user expectations.
- d) Measuring code coverage of a sorting algorithm.
- e) Testing database query performance from a user's perspective

4. a) A software development team has completed the requirement analysis and design phases of a project.

5 C02

- I. List and describe the major activities performed during implementation phase in SDLC.
- II. Discuss common challenges faced during implementation and how they can be minimized.

- b) A bank uses a **Banking Management System** for customer accounts, transactions, loans, and online services. The system requires continuous maintenance due to changing business requirements. Maintenance activities include:

5 C04

- I. Correcting transaction processing errors.
- II. Modifying the system to comply with new banking regulations.
- III. Enhancing security features and improving transaction speed.
- IV. Updating backup and recovery mechanisms.

Solve the following questions based on the scenario,

- a) Identify and explain the different types of software maintenance involved in scenario
- b) Explain why software maintenance is important after system deployment.
- c) Mention factors affecting the maintainability of the banking system.

OR (of 4b)

A regional bank is replacing its decades-old core banking system with a modern one. The bank has four branches: one small branch used mainly as a pilot site, and three large branches that handle high transaction volumes daily. Management is nervous about disrupting live customer transactions, wants to catch any bugs early with minimal risk, and would prefer not to run both systems fully in parallel everywhere due to the cost of double data entry. They are also considering rolling out functional modules (e.g., savings accounts first, then loans, then fixed deposits) one at a time rather than launching everything together.

Based on the four installation approaches (Direct, Parallel, Single Location, Phased), recommend a combined installation strategy for the bank, explaining which approach(es) apply at which stage and why.

- | | | | | |
|----|----|--|---|-----|
| 5. | a) | Discuss hardware and software selection criteria. | 5 | C02 |
| | b) | Why security is important in software implementation? Differentiate between the four kinds of security threats: interruption, interception, modification, and fabrication. | 5 | C03 |

OR (of 5b)

A small accounting firm's office has the following setup: employees write their system passwords on sticky notes attached to their monitors; the office network has no firewall or antivirus software installed; the office door is often left unlocked during lunch breaks; and last month, a former employee's login credentials were still active a week after they left the company. One afternoon, an unauthorized person walks into the office, uses a still-active former employee's password to log in, and copies confidential client financial data onto a USB drive. Using the concepts of Vulnerability, Exposure, Threat, and Control from your notes, identify each element present in this incident, and then classify the attack itself as one of the four kinds of security threats (interruption, interception, modification, fabrication).

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 International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring-2026
Course Code: EEE-2421 Course Title: Electrical Drives and Instrumentation
Total marks: 50 Time: 2 hours 30 minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Write down the advantages of a single-phase induction motor. Why is it not self-starting? Describe its starting mechanism with neat diagrams.	1	U	5
1.b	Explain the working principle of a variable frequency drive with its block diagram.	3	C	5
2.a	Briefly analyze the methods of making a synchronous motor self-starting and its applications.	1	An	5
2.b	What is micro-stepping? Explain the construction of a variable reluctance stepper motor and a permanent magnet stepper motor.	3	C	5
	OR			
2.a	How can you control the speed of a synchronous motor using rectifiers and inverters? Describe with figures.	1	An	5
2.b	What is stepper motor? Suppose a variable reluctance stepper motor has 8 stator teeth and 6 rotor teeth. Calculate the step angle.	3	C	5

Group B

Q. No.	Question	CLO	BT	Marks
3.a	What is meant by sensitivity in measurement and instrumentation? Describe the process of measurement including all stages.	1	U	5
3.b	An $820\ \Omega$ resistance with an accuracy of $\pm 10\%$ carries a current of 10 mA. The current was measured by an analog ammeter on a 25mA range with an accuracy of $\pm 2\%$ of full scale. i.) Calculate the power dissipated in the resistor ii.) Determine the accuracy of the result.	2	E	5

4.a

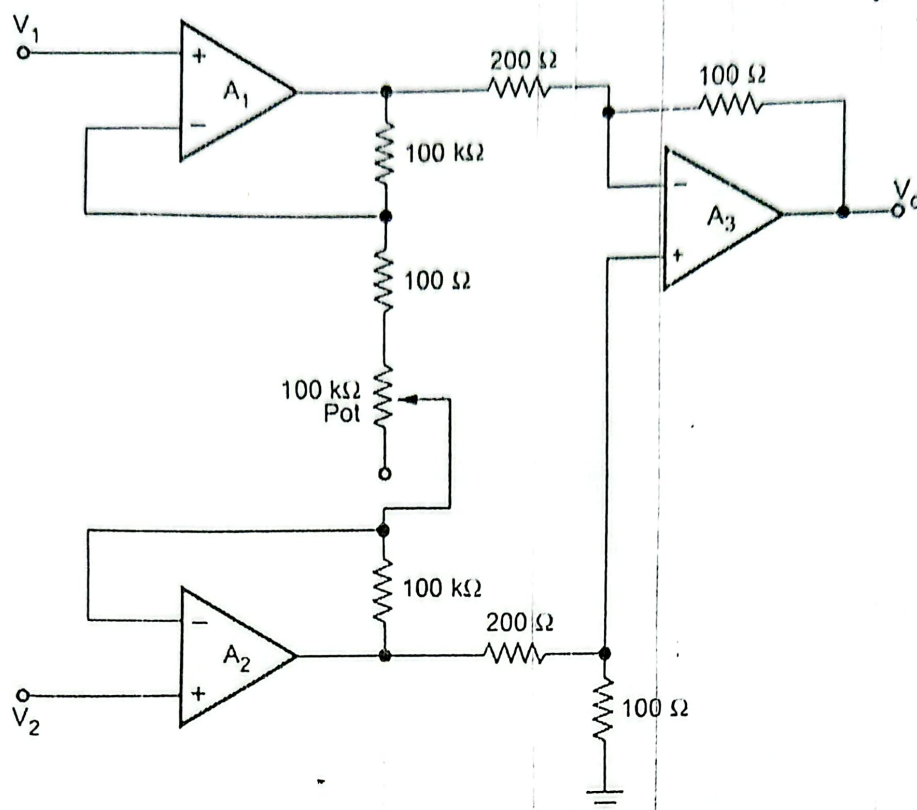


Fig.1

The circuit shown in Fig. 1 is an instrumentation amplifier. Determine the range over which its gain can be varied if the potentiometer is varied over its entire range.

4.b	Explain with the help of a functional block diagram, the operation of a digital frequency meter.	2	C	5
5.a	Analyze the basic principle of a Piezoelectric transducer and an ultrasonic piezoelectric transducer using the necessary diagram.	1	An	5
5.b	Write down the functions of a transducer. Also, describe the Resistance temperature detectors with a neat diagram.	2	U	5
OR				
5.a	Explain the working of Piezoelectric Transducer with proper circuit diagram.	1	An	5
5.b	Briefly describe the working principle of Thermocouple. During experiments with a copper-constantan thermocouple it was found that $c = 3.75 \times 10^{-2} \text{ mV}/^\circ\text{C}$ and $k = 4.5 \times 10^{-5} \text{ mV}/^\circ\text{C}^2$. If $T_1 = 100^\circ\text{C}$ and the cold junction T_2 is kept in ice, compute the resultant electromotive force.	2	U	5

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring-2026
Course Code: CSE 3521 Course Title: Computer Architecture
Total marks: 50 Time: 2.5 hours

Right column indicates: M = Marks, C = Course Outcome, D = Bloom's Taxonomy Domain.

Separate answer scripts must be used for Group A and Group B

Q.	GROUP-A	M	C	D
1(a)	Compare the execution of R-type, Load Word (LW), and Branch Equal (BEQ) instructions in a single-cycle MIPS processor. Using a single-cycle datapath diagram, highlight the data path and explain the control signals for each instruction.	5	3	4
	OR			
1(a)	In a MIPS multicycle processor, the execution of an instruction is divided into multiple states. Explain the sequence of states involved in executing an R-type instruction, and describe the operation performed in each state.	5	3	4
1(b)	Illustrate the complete MIPS Single-Cycle Datapath with all major components and control paths clearly labeled. Using your diagram, explain the roles of the Program Counter (PC), Sign Extend unit, Shift Left 2 ($\ll 2$), ALU, Zero signal, Adder, and PCSrc multiplexer. Also justify why the branch offset is shifted left by 2 bits before being added to PC + 4.	5	3	3
	OR			
1(b)	Explain the purpose of the Instruction Register (IR), Memory Data Register (MDR), ALUOut, and A/B registers in the multicycle datapath. Illustrate the process with suitable MIPS instructions.	5	3	3
2(a)	The current PC is 0x00000100, and the instruction beq \$t0, \$t1, 4 is executed. Assume the immediate value is interpreted as a signed offset. If the branch is taken, calculate the values of PC + 4, the branch offset after the $\ll 2$ operation, and the final branch target address. Show your calculations.	5	3	3
2(b)	A single-cycle MIPS processor has the following functional unit delays: Instruction Memory = 250 ps, Register File Read = 120 ps, ALU = 180 ps, Data Memory = 250 ps, Register File Write = 120 ps, Multiplexer = 30 ps, Sign Extender = 20 ps, and Control Unit = 40 ps. Determine: (i) the clock cycle time for an R-type (ADD) instruction; (ii) the clock cycle time for a Load Word (LW) instruction; and (iii) the instruction that determines the minimum clock period of the processor.	5	2	3
	GROUP-B			
3(a)	A five-stage MIPS pipeline executes the following instructions: LW R1, 0(R2), ADD R3, R1, R4, SUB R5, R3, R6, AND R7, R5, R8. Identify all data hazards and explain how these hazards can be resolved.	5	3	4
	OR			
3(a)	A pipelined MIPS processor uses a single memory for both instructions and data. Explain how a structural hazard can occur during the execution of a Load Word (LW) instruction. Illustrate your answer with a suitable diagram and suggest one hardware-based solution to eliminate the hazard.	5	3	4
3(b)	Design an instruction forwarding (bypassing) unit for a five-stage MIPS pipeline. Illustrate the forwarding paths and explain how your design eliminates data hazards without introducing unnecessary stalls.	5	3	5
4(a)	Explain the concepts of cache associativity and replacement policy. Compare Direct-Mapped, Fully Associative, and Set-Associative caches with examples.	5	2	4
4(b)	A computer has a 32 KB direct-mapped cache with a block size of 64 bytes. Determine the tag, index, and offset bits for a 32-bit memory address.	5	2	3
	OR			
4(b)	A 2-way set-associative cache contains 8 sets. Determine the cache set for the following memory block addresses: 19, 26, 41, and 58. Show all calculations using the set-associative mapping rule. State whether any two blocks can reside in the same set simultaneously.	5	2	3
5(a)	Design an efficient virtual memory system for a computer with limited physical memory. Justify whether paging, segmentation, or a combination of both would be the most suitable memory management technique.	5	2	5
5(b)	The new CSEiac server is designed to support multiple high-speed storage and multiple low-speed I/O devices. Recommend the appropriate type of I/O channel for each category of devices and justify your choice.	5	2	4

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Final Examination, Spring 2026

Course Code: CSE-3527 Course Title: Compiler

Full Marks: 50 Time: 2 Hour 30 minutes

[Answer all of the following questions from Group A and B. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

- | No. | Group – A [2×10 = 20 Marks] | M | COs | DL |
|--|---|---|-----|----|
| 1. a) | Syntax Analysis: elucidate the basic error recovery strategies to error handling in compiler construction. | 2 | 3 | U |
| b) | Syntax Analysis: Consider the following grammar G_4 over the alphabet $\Sigma = \{a, b, c, d\}$.
$S \rightarrow Aa \quad A \rightarrow aA \mid b$
i) Provide the DFA transitions and state items of the LR(1) canonical collections. The first LR(1) DFA state item (I_0) already been provided in Fig. 2(b) for your reference. | 3 | 3 | An |
| <p>Or What is the purpose of the left factoring in a parser? Is there any left factoring possible in the following grammar G_2? If yes, generate the left-factored of that G_2 over the language $\Sigma = \{if, then, else, a, b\}$.</p> <p style="margin-left: 40px;"> $Stmt \rightarrow if\ Expr\ then\ Stmt \mid if\ Expr\ then\ Stmt\ else\ Stmt \mid a$
 $Expr \rightarrow b$ </p> | | | | |
| c) | Syntax Analysis: Consider the following CFG G_3 over the language $\Sigma = \{_, a-z, A-Z, 0-9\}$ for C programming identifier:
$I \rightarrow _A \mid LA \quad A \rightarrow _A \mid LA \mid DA \mid \epsilon \quad L \rightarrow a-z \mid A-Z \quad D \rightarrow 0-9$
i) Construct the FIRST and FOLLOW sets for each of the non-terminals in the above grammar
ii) Construct the predictive parsing table based on the above grammar G_3 .
iii) Is this grammar LL(1)? Explain. | 5 | 3 | C |
| 2. a) | Syntax Analysis:
Write the basic difference of all LR parsers in compiler construction and also contrast LR(0) and LR(1) item in LR parser. | 3 | 3 | U |
| b) | How many possible types of conflict in a LR parsing table construction? Explain the possible types of conflict in a LR parsing in the following grammar G_5 over the alphabet $\Sigma = \{a, b, c, d\}$ and answer the following questions i) to iii):
$S \rightarrow AB \quad A \rightarrow aA \mid \epsilon \quad B \rightarrow aB \mid \epsilon$
i) Provide the DFA transitions and state items of the LR(0) collections. The first LR(0) DFA state item (I_0) already been provided in Fig. 2(b) Or for your reference.
ii) Construct LR(0) parser table if possible. The parser table includes both the action and goto tables.
iii) Show that above CFG G_5 is also not SLR(1) grammar. | 7 | 3 | C |
| <p>Or, Consider the following grammar:</p> <p style="margin-left: 40px;"> $S \rightarrow Sa \mid aA \mid aB \quad A \rightarrow cA \mid d \quad B \rightarrow cB \mid e$ </p> <p>Do the following task for the above grammar:</p> <p>i) Remove left recursion. ii) Apply left factoring.
 iii) Compute FIRST(S), FIRST(A), FIRST(B).
 iv) Compute FOLLOW(S), FOLLOW(A), FOLLOW(B).
 v) Construct the predictive parsing table. vi) Check whether the grammar is LL(1).</p> | | | | |

3. a) Syntax directed translation: What syntax-directed translation? Consider for the following grammar G_6 over the language $\Sigma = \{int, float, long, id, ', '\}$ with every non-terminal has inherited attributes like *.in* and *.type* for data type declaration [id representation of all identifiers for the C variables];

$$D \rightarrow TL$$

$$T \rightarrow int \mid float \mid long$$

$$L \rightarrow L, id \mid id$$

Give the syntax-directed definition of the following grammar G_5 . Construct the dependency graph with related instructions to declare the input statement string "long a, b, c, d"

- Or, Syntax directed translation: Explain inherited attributes with an example. Using the following SDT draw the annotated parse tree and dependency graph for the string $3*5*7$.

PRODUCTION	SEMANTIC RULES
1) $T' \rightarrow F T'$	$T'.inh = F.val$ $T.val = T'.syn$
2) $T' \rightarrow * F T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3) $T' \rightarrow \epsilon$	$T'.syn = T'.inh$
4) $F \rightarrow digit$	$F.val = digit.lexval$

- b) Type checking:

- Write the basic purpose of static checkers and type checking step in compiler construction?
- Write syntax-directed translation (semantic rules) for the following type checking statements syntax definitions:
 $S \rightarrow E (E)$ [for C function prototype definition]
 $S \rightarrow E [E]$ [for C array declaration]
 $E \rightarrow op E$ [for declaration of all C arithmetic operators]

4. a) Run-time environment:

- What is the activation tree in the run-time environment of a compiler for a function?
- What is the purpose of activation tree in the run-time environment of a compiler. Draw the activation tree corresponding to the following execution trace of quicksort:

```

Execution begins...
enter readarray
leave readarray
enter quicksort(1,9)
enter partition(1,9)
leave partition(1,9)
enter quicksort(1,3)
...
leave quicksort(1,3)
enter quicksort(5,9)
...
leave quicksort(5,9)
leave quicksort(1,9)
Execution terminated.

```


Or, Translate the following expression to quadruple, triple, indirect triple and polish notation: $a + b * c / c + b * c$

5 4 Ap

b) Intermediate code generation: Consider the following C program:

5 5 Ap

```
int main(){
    int a, b, c, d;
    scanf("%d%d%d", b,c,d);
    a=a+(b+c-1)*(b+c)+d; // '-' is unary minus
    print("%d", a);
    return 0;
}
```

Translate the C assignment statement into:

- i) Syntax tree
- ii) Three-address code
- iii) Triples
- iv) DAG
- v) Quadruples

5. a) Code generation and optimization:

6 5 E

Consider the following C function of merging in merge sort algorithm and answer the questions (i) to (iii):

```
void insertionSort(int arr[], int n) {
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}
```

- i) Translate the C programming code into three-address code.
- ii) Identify the basic block in three-address code.
- iii) Construct the flow graph from the three-address code.

b) Construct the DAG for the following expression:

4 5 An

$((a + a) + (a + a)) + ((a + a) + (a + a))$

International Islamic University Chittagong
Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Spring-2026

Course Code: CSE-3523 Course Title: Microprocessors, Microcontrollers and Embedded Systems

Total marks: 50 Time: 2 hours 30 minutes

Right column indicates: M = Marks, C = Course Outcome, D = Bloom's Taxonomy Domain.

Separate answer scripts must be used for Group A and Group B.

		Group-A	M	C	D
1.	a)	Define ISR and NMI. Compare hardware and software interrupts with suitable examples or application scenarios.	4	2	An
	b)	What is the Interrupt Vector Table (IVT)? Explain the interrupt handling process of the 8086 microprocessor.	6	2	U
		OR			
	b)	The 8086 microprocessor has only one INTR (Interrupt Request) pin. However, a practical computer system contains multiple interrupt-generating devices such as the keyboard, mouse, disk controller, and printer. How can all these devices be connected to the single INTR pin? Explain a suitable method with a neat diagram.	6	2	An
2.	a)	Define pipeline hazards. Compare pipelined and non-pipelined instruction execution. Explain how floating-point operations are executed in an arithmetic pipeline.	6	2	Ap
	b)	Discuss the different types of specialized microprocessors. Explain their working principles, key features, practical applications, and suitable examples.	4	2	An
		OR			
	b)	Define context switching. Explain the characteristics that distinguish an embedded microprocessor from a general-purpose microprocessor.	4	2	An
		Group-B			
3.	a)	Why is address decoding necessary in memory interfacing? Compare isolated I/O and memory-mapped I/O.	4	3	An
		OR			
	a)	Describe, with a timing diagram, how Mode 1 handles output handshaking using the control signals WR, ACK, INTR, and OBF.	4	3	An
	b)	Explain the architecture and operation of the 82C55 Programmable Peripheral Interface. Discuss the operating modes of the 82C55.	6	3	An
4.	a)	Explain the key features of a microcontroller. Differentiate between a microcontroller and an embedded system.	4	3	An
	b)	Draw and explain the architectural block diagram of the 8051 microcontroller.	6	3	U
		OR			
	b)	Design a Finite State Machine (FSM) for an Automated Teller Machine (ATM) that supports the following operations: card insertion, PIN verification, balance inquiry, cash withdrawal, cash deposit, fund transfer, and transaction cancellation. Clearly define all states and state transitions, and illustrate the FSM using a state diagram.	6	3	C
5.	a)	Discuss the major privacy risks in Internet of Things (IoT) systems. Explain the key privacy principles used to protect user data.	5	3	An
	b)	What is the Internet of Things (IoT)? Illustrate the three-layer architecture of the Internet of Things (IoT).	5	3	U